



JOB SAFETY ANALYSIS (JSA)

Wet Utilities Installation — Fire Water

PROJECT NAME:	[Placeholder]	DATE:	[Placeholder]
PROJECT #:	[Placeholder]	PHASE:	[Placeholder]
WORK ACTIVITY:	Fire Water / Fire Protection Main Installation — Trench Excavation, Pressure Pipe Laying, Hydrants, Valves, Thrust Restraint, Flushing & Testing		PREPARED BY: Safety Manager
COMPANY:	Wicker Construction Inc.	JSA TYPE:	Wet Utility Installation

SCOPE OF WORK

This JSA covers all wet-utility installation activities for the Fire Water system, including: trenching and excavation, dewatering, bedding, pipe installation, structure/valve/hydrant setting, tie-ins, testing, backfill, and surface restoration. It is written for field execution and must be reviewed with the crew before work begins.

This JSA is one of five (5) wet-utility JSAs: Storm Sewer, Sanitary Sewer, Industrial Waste / NCCW / INDW, Domestic Water, and Fire Water.

SCHEDULE

MILESTONE	DATE	DESCRIPTION / POTENTIAL IMPACTS
Predecessor Task	[Date]	Site clearing, erosion control, and access road preparation complete.
Start Date	[Date]	Begin trench excavation at connection / first valve. Maintain separation from sewer & non-potable lines.
Milestone #1	[Date]	First segment installed; valves and hydrant lead set and surveyed. Thrust restraint in place.
Milestone #2	[Date]	Trench advanced to mid-run. Dewatering operational if groundwater encountered.
Milestone #3	[Date]	All joints complete and restrained. Hydrostatic test (200 psi / spec) and flush scheduled and passed per NFPA 24.
Milestone #4	[Date]	Hydrants set plumb to grade, with thrust restraint and gravel drainage. Tracer wire installed and tested.
Milestone #5	[Date]	Final backfill and compaction complete. Compaction test reports submitted. Contractor's Material & Test Certificate completed.
End Date	[Date]	Punch-list resolved. As-builts submitted. Site restored. Note $\pm 10\%$ measurement accuracy where pipe schedule absent.
Successor Task	[Date]	Surface restoration, paving, or tie-in to existing fire main / building fire service / FDC & riser.

CREW & CONTACTS		
TRADE PARTNER CONTACTS	NAME	PHONE NUMBER
FOREMAN:	[Name]	[Phone]
SUPERINTENDENT:	[Name]	[Phone]
SAFETY PROFESSIONAL:	[Name]	[Phone]
WICKER SUPERINTENDENT:	[Name]	[Phone]
WICKER SAFETY PROFESSIONAL:	[Name]	[Phone]
CREW SPONSOR:	[Name]	[Phone]

REQUIRED CREW	PRODUCTIVITY REQUIREMENTS	WORK HOURS
• 1 × Foreman • 1 × Superintendent • 2 × Equipment Operators (Excavator) • 1 × Equipment Operator (Loader/Compactor) • 2 × Laborers (Trench) • 1 × Pipe Layer / Grade Setter • 1 × Truck Driver (Spoils Haul-off) • 1 × Spotter / Flagger • 1 × Safety Observer (excavation > 4 ft)	• Pipe Installation: 100–180 LF/day • Hydrant / Valve Setting: Per plan • Backfill & Compaction: Per lift per spec • Compaction Tests: Every 200 LF and each lift • Hydro / Flush Test: Per NFPA 24 & spec	• Standard: [Start] – [End] • Monday – Friday • Saturday as required • Overtime subject to owner approval • No work during lightning standdown

MATERIALS		
MATERIAL	QTY	NOTES / STORAGE REQUIREMENTS
Fire Water Pipe (DI / PVC C900 / HDPE per plan & class)	Per plan LF	Verify pressure class for fire service. Keep interior clean and capped. Protect PVC from UV.
Fire Hydrants (per plan / AWWA C502)	Per plan EA	Verify barrel length / bury depth to grade. Inspect operation before install.
Valves (gate / post indicator, per plan) & Valve Boxes	Per plan EA	Verify size/class. PIV where required. Set valve box plumb to grade.
Fittings (bends, tees, reducers) & Joint Restraints	Per plan EA	Restrain per thrust schedule. Verify pressure class matches main.
Crushed Stone Bedding & Hydrant Drainage Gravel	Per plan TN	Stockpile in laydown area. Place drainage gravel at hydrant base per detail.
Select Backfill (native or imported per spec)	Per plan CY	Verify approval with geotechnical engineer before placing.
Gaskets / Lubricant (mfr. supplied)	Per joint	Use only manufacturer-approved lubricant. Keep shaded. Do not freeze.
Thrust Blocking Concrete / Mechanical Restraint	As needed	Per thrust detail. Allow cure before pressure test. Critical at hydrant leads & bends.
Tracer Wire & Marking Tape (per local standard)	Per plan LF	Continuous; tested for continuity.
Hydrant Markers / Protection Posts (if specified)	Per plan EA	Install per plan to protect hydrants from traffic.

TOOLS / EQUIPMENT

TOOL / EQUIPMENT	QTY	NOTES
Excavator — Long Reach (Cat 320 or equiv., trench bucket)	1	Inspect boom, bucket teeth, and hydraulic lines daily.
Excavator — Standard (Cat 308 or equiv., for backfill / setting)	1	Used for precise backfill placement and trench box advancement.
Steel Trench Box / Trench Shield (rated for excavation depth)	1 set	Inspect spreaders and wales daily. Competent Person required.
Vibratory Plate Compactor (walk-behind, gas)	2	Use for compacting lifts adjacent to pipe. Do not compact directly on pipe.
Dewatering Pump (submersible, 3" or 4" discharge)	2	Discharge per permit — no direct surface discharge without approval.
Pipe Laser / Grade Laser (Topcon or equiv.)	1	Calibrate daily. Verify against benchmark.
Certified Lifting Bail / Slings (rated for heaviest pick)	2	Inspect before each lift. Match rating to heaviest section / pipe.
Confined Space Entry Kit (4-gas monitor, tripod, SRL, harness, winch)	1 set	Required when personnel enter any structure or excavation >4 ft. Calibrate daily.
Wheel Loader (for bedding / backfill material staging)	1	Used to move material from stockpile to trench area.
Dump Trucks (spoils haul-off)	2–3	Coordinate with traffic control. Cover loads leaving site.

HIGH-RISK ACTIVITY / ERROR-LIKELY

Attach specialized plans to the end of this document

HIGH-RISK ACTIVITY	SPECIAL PLAN REQUIRED?
Excavation / Trenching exceeding 5 ft depth — Cave-in / Engulfment hazard	YES ☒ NO ☐
Stored pressure energy during hydrostatic testing (200 psi / spec) — blow-off & thrust movement	YES ☒ NO ☐
Heavy hydrant / valve handling — struck-by / crush	YES ☒ NO ☐
Thrust failure at hydrant leads, bends, and dead ends	YES ☒ NO ☐
Dewatering Operations — Groundwater infiltration, trench instability, discharge permit compliance	YES ☒ NO ☐

REQUIRED PERMITS

REQ'D	PERMIT / AUTHORIZATION
☐	Daily Dig / Excavation Permit (signed by Wicker Superintendent before any soil disruption)
☐	Confined Space Entry Permit (any structure or excavation >4 ft)
☐	Hot Work Permit (cutting / welding / grinding — if applicable)
☐	Dewatering / Stormwater Discharge Permit ([Permit No. / Agency])

☐	Line Breaking / Energy Isolation (LOTO) Permit for tie-ins to live systems
☐	Locate / One-Call (811) Ticket — current and valid before excavation
☐	Crane / Critical Lift Permit (if pick exceeds standard rigging plan)
☐	Other (project-specific): _____

EMERGENCY RESPONSE & STOP-WORK AUTHORITY

STOP-WORK AUTHORITY: Every worker has the unconditional right and duty to STOP WORK for any unsafe condition — trench instability, atmospheric alarm, utility strike, rigging concern, equipment fault, weather, or any uncertainty. Work does not resume until the condition is corrected and the Competent Person / Wicker Superintendent authorizes restart. No retaliation. Zero-incident mindset.

EMERGENCY PROCEDURES:

- Call 911. State exact site location / address: [Enter address & gate]. Send a runner to direct EMS.
- Notify Wicker Superintendent and Wicker Safety immediately: [Phone].
- Muster Point: [Enter location]. Account for all personnel. Report headcount to Superintendent.
- First Aid / AED location: [Enter]. Nearest hospital: [Enter name, address, phone].
- Trench collapse / engulfment: Do NOT enter to rescue. Call 911 / trained rescue. Shut down equipment, secure area.
- Atmospheric alarm or confined-space emergency: Entrant exits; attendant initiates non-entry rescue; never enter to rescue.
- Utility strike (gas/electric/comm): Stop work, evacuate, keep clear, call 911 and the utility owner. No equipment movement.
- Spill / release: Stop source if safe, deploy spill kit / containment, report to Wicker Safety and per permit.
- Severe weather / lightning: Standdown at 15 miles; resume 30 min after last strike. Exit trench.

CONTROLS REFERENCE — ENGINEERING · ADMINISTRATIVE · PPE

Applies to all tasks below. Task-specific controls are listed in the JHA.

ENGINEERING CONTROLS	ADMINISTRATIVE CONTROLS	PPE (MINIMUM)
• Trench box / shield, sloping, or shoring (>5 ft) • Spoils set back ≥4 ft from edge • Mechanical ventilation in confined spaces • Dewatering pumps & sumps • Ladder access within 25 ft, 3 ft above edge • Tag lines, restraints & thrust blocking • Trench plates / covers for after-hours • Backup alarms, cameras on equipment	• Competent Person daily trench inspection • Daily Dig & Confined Space permits • 811 locate; pothole; hand-dig within 3 ft • Pre-task huddle / JSA review & sign-on • 4-gas monitor calibrated & continuous • Trained/qualified operators & riggers • 15-ft exclusion zone & signage • Heat illness & lightning standdown plans	• Hard hat (ANSI Z89.1) • Hi-vis Class 2/3 vest • Safety glasses; face shield as needed • Steel/composite-toe boots • Cut-resistant (A4) gloves • Hearing protection near equipment • Harness/SRL near open edge & CSE • Task PPE: chem-resistant gloves/suit, FR, respirator per SDS/task

QUALITY

Required inspections & frequency, critical tolerances, and key performance indicators.

ITEM	REQUIREMENT
1. Separation from Other Utilities	Maintain code-required separation from sewer / non-potable lines. Pothole and verify at crossings. Confirm cover/bury per plan and freeze depth.
2. Pipe Depth, Grade & Alignment	Verify depth of cover per plan. Alignment per plan. Keep pipe interior clean and capped when not laying.
3. Bedding & Initial Backfill	Bedding to springline before joint. No rocks/frozen material within 12" of pipe. Haunch per material.
4. Thrust Restraint (critical)	All fittings, hydrant leads, valves, bends, and dead ends restrained per thrust schedule. Thrust blocks cured before test. This is the controlling quality item for fire mains.
5. Hydrants	Set plumb to finish grade. Drainage gravel at base per detail (unless prohibited). Operate to verify. Aux valve on hydrant lead.
6. Test, Flush & Certify	Hydrostatic test per NFPA 24 (typ. 200 psi / spec) and flush to required velocity. Complete Contractor's Material & Test Certificate. Document and submit.

REFERENCE DOCUMENTS & DRAWINGS

1. Approved Construction Drawings — Fire Water / Fire Protection Plan & Profile Sheets (Sheet Nos. [X] through [X]); hydrant / valve / thrust schedules
2. Project Specifications — Division 33: Section 33 11 00 / Fire Service Mains
3. NFPA 24 — Installation of Private Fire Service Mains; AWWA C600 / C605, C502 (hydrants), C900 (PVC) as applicable
4. Geotechnical Report — Soils Investigation & Recommendations — [Geotechnical Firm, Date]
5. Engineer of Record — Bohler Engineering — applicable fire-service details and standards (TxDOT / NTCOG as specified)
6. OSHA 29 CFR 1926 Subpart P — Excavations (Trenching & Shoring)
7. Wicker Construction Inc. — Site-Specific Health & Safety Plan (HASP)
8. Wicker Construction Inc. — Excavation & Trenching Safety Plan
9. Dewatering & Discharge Permit — [Permit No. / Agency]
10. Pipe / Hydrant Manufacturer Installation Manual — [Manufacturer Name]

JOB HAZARD ANALYSIS (JHA) — FIRE WATER MAIN INSTALLATION

OPERATIONAL STEPS IN LOGICAL ORDER. FOLLOW IN THE FIELD AS AN INSTRUCTION MANUAL: SETUP, OPERATION, INSPECTION, STOPPING POINTS, CLEANUP.

#	WALK-THROUGH STEP	HAZARDS	RISK	CONTROLS & SAFE PROCEDURES
1	All personnel informed of emergency procedures prior to performing any work on site.	- Worker injuries and exposures. - Hazardous flora and fauna. - Temperature extremes / heat stress.	LOW	- Emergency: Call 911. Know exact site location and address. Designated Muster Point staffed. Notify Wicker Safety immediately. - First Aid Kit location: [Enter Location]. Nearest hospital / urgent care posted: [Enter]. - All employees complete Site-Specific Safety Orientation before any work begins. - Flora/Fauna: Know poison ivy/oak/sumac, fire ants, snakes. Threatening animal — stop work, retreat, notify Supervisor and Safety. - Heat Stress: Water provided. Breaks in shade / A-C. Reference Wicker Heat Illness Prevention Plan. Buddy system in extreme heat.
2	Mobilize equipment (excavators, loader, trench box, compactor, dewatering pumps) to laydown yard via access road.	- Equipment/vehicle collisions on access road. - Poor road conditions or obstructions. - Dust impairing visibility. - Tipping/overturning on unstable ground.	MED	- Spotters and radio communication between all operators. - Inspect access road for stability and obstructions before mobilization. - Apply water / dust control as needed. - Keep delivery trucks centered — away from ditch and excavation edges. - Verify equipment weight does not exceed road / mat capacity.
3	Unload and stage equipment in designated laydown area. Equipment fueled prior to arriving on site.	- Slips, trips, falls on uneven ground. - Reduced visibility / blind spots during unloading. - Unauthorized personnel entering unloading zone. - Pinch points during chain/strap removal.	LOW	- Only trained / qualified personnel operate mobile equipment. - Flaggers / spotters required for every delivery. - Controlled access zone established opposite trailer. - Cut Level A4 gloves when unhooking safety chains. - Team lift required for items over 50 lbs or longer than 10 ft.
4	Superintendents / Forepersons inspect all equipment and work areas. Verify existing utilities. Complete Daily Dig Permit.	- Faulty or damaged equipment. - Exposure to moving equipment during inspection. - Slips, trips, falls. - Hydraulic leaks / spills. - Existing underground utilities struck.	MED	- Daily equipment maintenance check sheet completed and submitted to Wicker Superintendent. - GIS / One-Call (811) tickets current — all utilities marked and flagged before excavation.

				• Daily Dig Permit signed by Wicker Superintendent before any soil disruption. • LUAWA completed for excavation within 10 ft of any live utility. • No mechanical digging within 3 ft of known live utility — hand dig / pothole required. • Spill kits on site. Report spills immediately. • Lightning standdown at 15 miles. 30-min standdown from last strike.
5	Set up 15-ft exclusion zone around excavation area with weighted cones, red barrier chain, and signage. Establish Competent Person inspection protocol.	• Unauthorized personnel entering excavation zone. • Trench box not inspected before use. • Unidentified underground utilities in excavation path.	MED	• Delineate 15-ft exclusion zone. Post 'Excavation in Progress — Keep Out' signage. • Competent Person (CP) completes written daily excavation inspection BEFORE any worker enters trench. • All utilities physically potholed and located before excavator begins digging. • Field Engineer stakes pipe alignment and structure locations before excavation starts. • Daily Dig Permit signed and posted at excavation before work begins.
6	Begin trench excavation. Set spoils minimum 4 ft from edge of trench. Competent Person inspects trench before worker entry.	• Cave-in / trench wall collapse — engulfment hazard. • Workers struck by excavator or falling spoils. • Soil instability due to groundwater or poor soil type. • Equipment tipping from spoils surcharge near trench edge.	HIGH	• Competent Person classifies soil per OSHA 29 CFR 1926 Subpart P before excavation. • All excavations >5 ft require trench box, sloping, or shoring before worker entry. • Spoils minimum 4 ft from trench edge — no exceptions. • No workers in trench while excavator is actively digging. • Dewatering pump deployed if groundwater reaches excavation bottom. • Atmospheric monitoring required for all excavations >4 ft — test O₂, H₂S, CO, LEL before every entry.
7	Install trench box / trench shield. Establish fall protection, OSHA-compliant ladder access, and tie-off points before any workers enter excavation.	• Workers entering unshored excavation. • Trench box failure or improper installation. • Falls into open excavation. • Workers struck by trench box during installation.	MED	• Trench box inspected by Competent Person before installation — document inspection. • No worker entry until trench box is properly set and CP approves entry. • OSHA-compliant ladder extending 3 ft above trench edge installed within 25 ft of all workers. • SRL attached to certified tie-off point when near open excavation edge. • Laborers clear of area when trench box is being set or advanced.

				• Maintain spoils minimum 4 ft from trench edge at all times.
8	Prepare pipe bedding: place and compact bedding stone to required depth below pipe invert. Field Engineer verifies grade before pipe is set.	• Improper bedding causing pipe settlement or joint failure. • Workers struck by equipment placing bedding. • Grade error resulting in non-compliant pipe slope / cover.	MED	• Place bedding below pipe invert per spec. Set to laser / plan grade. • Field Engineer checks trench bottom elevation before any bedding is placed. • Workers in trench maintain visual and radio contact with excavator operator at all times. • Excavator operator uses slow, deliberate movements when placing bedding. • Pipe laser verified against established benchmark before bedding begins.
9	Set fire hydrants and valves with thrust restraint and base drainage gravel; set plumb to finish grade.	• Struck-by / crush during heavy hydrant & valve handling. • Hydrant tip-over before backfill / restraint. • Inadequate thrust restraint at hydrant lead.	MED	• Rig hydrants/valves with rated slings; tag lines; keep workers clear of suspended/seated load until stable. • Brace hydrant and install restraint before releasing rigging. Do not leave unrestrained hydrant standing in open trench. • Install and cure thrust restraint at hydrant lead, tee, and aux valve per detail before pressurization. • Place drainage gravel at hydrant base per detail. Set plumb and to grade.
10	Hydrostatic test (typ. 200 psi / spec) and flush the fire main. Verify all restraint before pressurizing.	• Stored hydraulic energy at higher fire-service test pressure — severe blow-off / projectile hazard. • Thrust movement at hydrants, bends, tees, dead ends. • High-velocity flush water — erosion, footing loss, injection injury.	HIGH	• Confirm ALL thrust blocks cured and restraints installed before test. Fire mains test at higher pressure — restraint verification is mandatory. • Clear all non-essential personnel from the test area. No one in line with hydrants, plugs, caps, or fittings during test. • Pressurize slowly in steps; bleed air; hold per NFPA 24 / spec. Inspect from a safe position. • Control flush discharge to a stable, permitted location. Never strike/tighten fittings under pressure; depressurize before adjustment.
11	Perform atmospheric monitoring and confined space entry (CSE) for joint inspection, grade confirmation, or structure work inside trench / structure.	• O₂ deficiency or toxic gas (H₂S, CO, methane) in trench or structure. • Engulfment if trench wall collapses during entry. • Worker unable to self-rescue from excavation. • Heat illness in confined, low-	HIGH	• Confined Space Entry Permit completed and posted before ANY worker enters excavation >4 ft or any structure. • 4-gas monitor (O₂, H₂S, CO, LEL) calibrated and tested daily. Test before entry and continuously during.

		airflow trench.		- Acceptable atmosphere: O₂ 19.5–23.5%, H₂S <10 ppm, CO <35 ppm, LEL <10%. - Rescue plan established. Tripod, SRL, and rescue winch at entry point. Attendant at surface — NO entry to rescue. - Minimum 2 workers for confined space entry: 1 entrant, 1 attendant. - Ventilation fan deployed to maintain fresh air in deep / enclosed sections. - If alarm activates — immediate exit. Do not re-enter until source identified and cleared.
12	Perform initial backfill from pipe springline to 12" above pipe crown using approved material. Compact in lifts per spec. Third-party compaction testing.	- Pipe deflection or damage from improper backfill or compaction. - Workers struck by equipment during backfill. - Trench box movement damaging pipe during advancement. - Compaction equipment too close to pipe.	MED	- Initial backfill placed by hand or small equipment — no heavy equipment directly over pipe until adequate cover (per spec, typ. 24"). - Walk-behind vibratory plate for initial lifts. No jumping-jack compactor directly over pipe. - Third-party geotechnical inspector performs proctor testing per spec — every 200 LF and each lift. - Workers clear of excavator backfill zone. Radio and eye contact maintained. - Advance trench box in lifts — never pull box while workers are in trench. - CP re-inspects excavation walls after trench box is advanced.
13	Continue excavation, pipe installation, and backfill sequencing (repeat prior steps) as work progresses throughout the day.	- Complacency from repetitive tasks. - Fatigue increasing risk of errors. - Changing soil conditions as excavation advances. - Dewatering failure allowing water accumulation in trench.	MED	- Daily morning huddle: review JSA, JHA, and lessons learned from prior shift. - CP re-inspects trench at start of each day and after any rain event. - Dewatering pumps monitored throughout shift — dedicate one laborer to monitor pump operation. - Rotate workers in trench positions to reduce fatigue. Max 2-hour continuous trench work before break. - Foreman maintains constant awareness of trench conditions — STOP WORK if soil behavior changes. - Any near-miss or incident reported to Wicker Safety immediately — document and review before resuming.
14	Complete final backfill above pipe zone in compacted lifts using approved material. Loader / dozer	- Uneven / insufficient compaction causing surface settlement. - Equipment collision — loader	MED	Clear radio communication between excavator and loader operators at all times.

	and plate compactor used for upper lifts.	and excavator in close proximity. - Workers struck by loader or compactor. - Compaction test failure requiring rework.		- Enforce no-walk zones around loader and compactor operation. - Minimum 1-ft clearance from placed pipe during compaction. - Third-party compaction testing at each lift per spec — do not advance to next lift until passing results received. - CP / Foreman signs off on each lift before proceeding.
15	End-of-day equipment refueling in designated fueling area. Secure all open trenches and structures before leaving site.	- Fire hazard during refueling. - Chemical spill — fuel or hydraulic fluid. - Public / unauthorized entry into open trench after hours.	MED	- All equipment completely shut off before refueling. No open flames within 50 ft. - Spill kit at fueling station. Fueling operator certified. Containment deployed under transfer hose. - Report all spills immediately. Document on spill report form. - Open excavations secured with trench plates, steel decking, or water-filled barriers after hours. - Warning lights and reflective signage placed around all excavations before leaving site. - Wicker Superintendent verifies site is secure before last person leaves.

RISK KEY: **HIGH** = serious injury/fatality potential — special plan & extra controls. **MED** = controllable with standard controls. **LOW** = routine.

KEY SITE-SPECIFIC HAZARDS & CONTROLS — Fire Water

HAZARD	RISK	PRIMARY CONTROLS
Cave-in / Engulfment (excavations >5 ft)	HIGH	CP daily inspection & soil classification; trench box/shield, sloping or shoring before entry; spoils ≥4 ft from edge; no entry while digging.
Confined Space (structures & deep trench)	HIGH	CSE permit; 4-gas monitor before & continuously; ventilation; tripod/SRL/winch; attendant — non-entry rescue only.
High-pressure hydro test (200 psi / spec)	HIGH	Verify ALL thrust restraint cured before test; clear test zone; step-pressurize & bleed air; no one in line with fittings/hydrants.
Struck-by / caught-between (equipment, loads)	MED	Spotters & radio; exclusion/no-walk zones; tag lines; certified rigging inspected each lift; never under suspended load.
Underground utility strike	HIGH	Current 811 ticket; pothole & hand-dig within 3 ft; LUAWA within 10 ft of live utility; stop/evacuate/call on strike.
Dewatering & discharge compliance	MED	Monitor pumps each shift; discharge per permit; no untreated/unpermitted surface discharge; treat as required.
Heat illness & weather	MED	Water/rest/shade; acclimatization; buddy system; lightning standdown at 15 mi, resume 30 min after last strike.

APPROVAL SIGNATURES

PREPARED BY — SAFETY MANAGER, Wicker Construction Inc. Signature: _____ Print Name: _____ Date: _____	WICKER SUPERINTENDENT Signature: _____ Print Name: _____ Date: _____
FOREMAN Signature: _____ Print Name: _____ Date: _____	WICKER SAFETY PROFESSIONAL Signature: _____ Print Name: _____ Date: _____
PROJECT MANAGER Signature: _____ Print Name: _____ Date: _____	OWNER / GC REPRESENTATIVE (as required) Signature: _____ Print Name: _____ Date: _____

CREW REVIEW & SIGN-ON: By signing the attached crew sign-on sheet, each worker confirms this JSA was reviewed in the daily huddle, they understand the hazards and controls, and they understand their Stop-Work Authority.

REVISION LOG

DATE	REV #	DESCRIPTION OF CHANGE	BY
[Date]	Rev 0 — Original	Initial JSA release — Fire Water wet-utility installation.	Safety Manager
